DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTING SCIENCE
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS

Second Semester, SY 2024-2025

COURSE CODE	**:**	**S-ITCS227**
COURSE TITLE	**:**	**APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES**
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two units
MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	None
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Ms. Erika Kyle D. Colocado
CONSULTATION	**:**	
CONTACT DETAILS	**:**	edcolocado@dlsud.edu.ph (MS Teams)

VISION:

A leading Catholic university that inspires excellence and drives innovation towards a just, peaceful, and sustainable society.

MISSION:

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension program.

COURSE DESCRIPTION:

This course focuses on the development of applications using web, mobile and emerging technologies with emphasis on requirements management, interface, design, usability, testing and deployment, including ethical and legal considerations.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. interpret the difference of front-end and back-development in web application;
CLO2. understand Web Development concept by dealing client-based case study;
CLO3. implement Object Oriented way of programming;
CLO4. manage website using Model, View and Control;
CLO5. develop and deploy complete Web Application in live server for pilot testing; and
CLO6. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

CLO1	Interpret the difference of front end and back-development in web application.	TLO1	Learn and differentiate different web development software and its features
CLO2	Understand Web Development concept by dealing client-based case study.	TLO2	Obtain tools in developing Web Application Project. Students can implement objects in their codes.
CLO3	Implement Object Oriented way of programming.	TLO3	Design web application both front-end and back-end. Students can know and use different OOP programming techniques.
CLO4	Manage website using Model, View and Control.	TLO4 TLO5	Design and Implements server-side controls. Design and Implement Master page.
CLO5	Develop and deploy complete Web Application in live server for pilot testing.	TLO6	Create and Design Web application report tools.
CLO6	Demonstrate 21st century skills in all learning activities.	TLO7	Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real- World Problem- Solving, Use of ICT (not part of topic, but integrated in assessments)

Module No. and Learning Outcomes	**Week/Date**	**Teaching-Learning Activities/Assessment Strategies**	**Allotted Hours**
Module 0	Week 1: Jan. 20 - 25	• **Gospel Reading, Reflection and Prayer** • **Course Orientation** ○ Course Syllabus ○ Course Overview/Orientation	3
Module 1 CLO1 CLO2 CLO4 CLO5 CLO6 TLO1 TLO2 TLO7	Week 2: Jan. 27 – Feb. 1	• **Gospel Reading, Reflection and Prayer** • **Module 1: Introduction to APIs** ○ Understanding APIs and their role in modern web applications • **Enabling Assessment: Laboratory 1** (Essay about understanding and comparing APIs)	3
Module 2 CLO1 CLO2 CLO4 CLO5	Week 3: Feb. 3 – Feb. 8	• **Gospel Reading, Reflection and Prayer** • **Module 2: Understanding APIs and web development**	3

CLO6 TLO1 TLO2 TLO6 TLO7		○ Introduction to client-side development (HTML, CSS, JavaScript) • **Enabling Assessment: Laboratory 2** (Making a simple web using HTML, CSS and JavaScript)	
Module 2 CLO1 CLO2 CLO4 CLO5 CLO6 TLO1 TLO2 TLO4 TLO7	Week 4: Feb. 10 – Feb.15	• **Gospel Reading, Reflection and Prayer** • **Module 2: Understanding APIs and web development** ○ Introduction to client-side development (HTML, CSS, JavaScript) ○ Understanding the client and server using APIs ○ Basic concepts of RESTful APIs ○ Creating RESTful APIs with CRUD operation (Create, Read, Update, Delete) • **Enabling Assessment: Laboratory 3** (Fetching data using public API to website)	3
Module 3 CLO3 CLO6 TLO1 TLO3	Week 5: Feb. 17 – Feb. 22	• **Gospel Reading, Reflection and Prayer** • **Module 3(Part 1): Backend Development** ○ Object-oriented programming concepts applied to server-side development ○ Understanding backend frameworks	3

TLO6 TLO7		• **Enabling Assessment: Laboratory 4** (Basic programming of Python using Object-oriented programming)	
Module 3 CLO3 CLO6 TLO1 TLO3 TLO6 TLO7	Week 6: Feb. 24 – Mar. 1	• **Gospel Reading, Reflection and Prayer** • **Module 3(Part 2): Backend Development** ○ Object-oriented programming concepts applied to server-side development ○ Understanding backend frameworks • **Enabling Assessment: Laboratory 5** (Continuation of coding the Python using Object-oriented programming)	3
Module 4 CLO1 CLO2 CLO6 TLO1 TLO2 TLO6 TLO7	Week 7: Mar. 3 – Mar. 8	• **Gospel Reading, Reflection and Prayer** • **Module 4: Master Pages** • **Enabling Assessment: Laboratory 6** (Design a master page layout to maintain a consistent structure across multiple web pages)	3
		• **Gospel Reading, Reflection and Prayer** • **Review for the midterms**	3

	Week 8: Mar. 10 – Mar. 15		
	Week 9: Mar. 17 – Mar. 22	**MIDTERM SUMMATIVE ASSESSMENT** • Creating an application using HTML, CSS, JavaScript and Python	3
			27

Learning outcomes of Final Period:

CLO5	Create a Python application that accesses an API based on user input and processes and displays the JSON data that is returned.	TLO5	Understand and learn APIs and JSON.
CLO7 CLO8	Evaluate the importance of working with your instructor and fellow students to make the material in this course applicable to the local situation. Demonstrate 21st-century skills in all learning activities.	TLO7	Application of the course material to local situations. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of the topic, but integrated into assessments)

Module No. and Learning Outcomes	**Week/Date**	**Teaching-Learning Activities/Assessment Strategies**	**Allotted Hours**
Module 5		• Gospel Reading, Reflection and Prayer	2.25

CLO5 CLO7 CLO8 TLO5 TLO7	Week 10: Mar. 24 – Mar. 29	• **Module 5: Introduction to Database** ○ Databases concepts and connectivity • **Enabling Assessment: Laboratory 7** (Essay about the understanding of databases)	
Module 5 CLO5 CLO7 CLO8 TLO5 TLO7	Week 11: Mar. 31 – Apr. 5	• Gospel Reading, Reflection and Prayer • **Module 5: Introduction to Databases** • **Enabling Assessment: Laboratory 7** (Inputting data in the database, Understanding the basics MySQL)	2.25
Module 8 CLO5 CLO7 CLO8 TLO5 TLO7	Week 12: Apr. 7 – Apr. 12	• Gospel Reading, Reflection and Prayer • **Module 6: Understanding the real-time applications** • **Enabling Assessment: Laboratory 8** (Essay about understanding and comparing the real-time applications) (Making their own application for Final summative)	2.25
	Week 13:	Independent Learning Days	2.25

	Apr. 14 – Apr. 19	(Apr. 14 – Apr. 15) Holy Week (Apr. 16 – Apr. 19)	
	Week 14: Apr. 21 – Apr. 26	Research week	2.25
	Week 15: Apr. 28 – May 3	• **Module 7 (Part 1): Emerging technologies** ○ Understand how technologies emerging • **(Making their own application for Final summative)**	2.25
Module 8 CLO5 CLO7 CLO8 TLO5 TLO7	Week 16: May 5 – May 10	• Gospel Reading, Reflection and Prayer • **Module 7(Part 2): Emerging technologies** ○ Emerging technologies concepts and features • (Making their own application for Final summative)	2.25
	Week 17: May 12 - 17	• Review week in preparation for the Final Examination • Completion of assessments/requirements	2.25

	Week 18: May 13 – May 24	**FINALS SUMMATIVE ASSESSMENT** Presentation of their Final application	2.25
			20.25

GRADING SYSTEM:

Midterm	**%**	**Final**	**%**
Enabling Assessments	40%	Enabling Assessments	40%
Summative Assessment (Midterm Examination)	60%	Summative Assessment (Final Examination)	60%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies

a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.

b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.

d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback

a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:

i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.

ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.

iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy

a. Attendance Policy for Onsite Classes is scheduled based on the learning plan.

b. The onsite meeting will only be held at the schedule indicated in the registration form.

c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy

a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.

b. Using any illegally obtained software is strictly prohibited.
c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies

a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES:

Online References:

Call number/ E-provider	Reference Material
e-web.medium	Byers, N. Understanding APIs: A comprehensive guide to integration in web development. *Medium*. Retrieved from https://medium.com/@nathanbyers13/understanding-apis-a-comprehensive-guide-to-integration-in-web-development-7241bb3ad793
e-web.appmaster	AppMaster. APIs in web development. *AppMaster Blog*. Retrieved from https://appmaster.io/blog/apis-in-web-development
e-web.freecodecamp	Ubah, K. Learn web development basics – HTML, CSS, and JavaScript explained for beginners. *freeCodeCamp*. Retrieved from https://www.freecodecamp.org/news/html-css-and-javascript-explained-for-beginners/

e-web.apidog	Apidog. Client-side APIs vs. server-side APIs: A clear comparison. *Apidog Blog*. Retrieved from https://apidog.com/blog/api-client-side-vs-api-server-side/
e-web.geeksforgeeks	GeeksforGeeks. How to use an API? The complete guide. *GeeksforGeeks*. Retrieved from https://www.geeksforgeeks.org/how-to-use-an-api-the-complete-guide/
e-web.redshift	Meyer, A. Mastering backend development: Fundamental concepts and terminologies. *redShift Recruiting Blog*. Retrieved from https://www.redshiftrecruiting.com/career-blog/mastering-backend-development-fundamental-concepts-and-terminologies
e-web.geeksforgeeks	GeeksforGeeks. REST API using the Express to perform CRUD (Create, Read, Update, Delete). *GeeksforGeeks*. Retrieved from https://www.geeksforgeeks.org/rest-api-using-the-express-to-perform-crud-create-read-update-delete/
e-web.techtraininghq	Tech Training HQ. What is emerging technology? *Tech Training HQ*. Retrieved from https://www.techtraininghq.com/what-is-emerging-technology/
e-web.startusinsights	StartUs Insights. Emerging technologies full guide. *StartUs Insights*. Retrieved from https://www.startus-insights.com/innovators-guide/emerging-technologies-full-guide/

Prepared: Computer Science Department

2nd Semester 2024-2025

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